

Digital Research Toolkit for Linguists

Week 6: Data visualization

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May 13, 2025

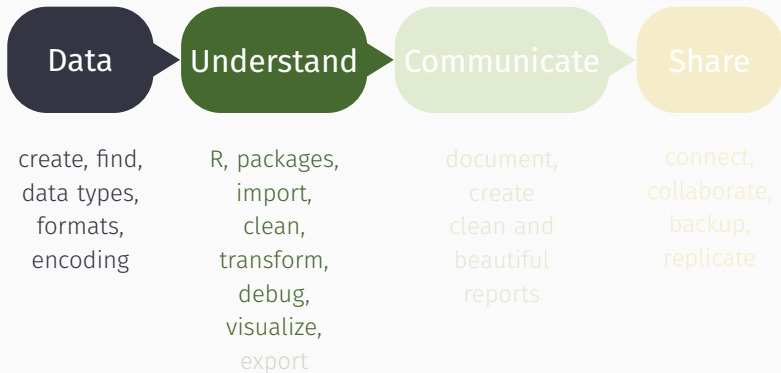
Psycholinguistics and Cognitive Modeling Lab

Homework

Questions?

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Tidy code

A rose by any other name

Names **uniquely identify** variables and functions

Capitalization matters

`DATA != Data != data`

Everyone has an opinion/preference

- ✓ `for_snake_case`
- ✓ `somePreferCamelCase`
- ✓ `others.use.periods`
- ✓ `Some_people.just.WANT_To_Watch.theWorldBURN`
- ✓ `Num8er5` are fine
- ✗ `Sp aces` are not
- ✗ `Speci@l+character$&operator<s>!` are a illegal

R has no official style guide, but preference for alphanumeric & _

CALL YOU?



**I'VE ALREADY FORGOTTEN YOUR
NAME**

R doesn't care but you do

✓ `ue0d2FNRGAP0dRopq40qU`

✓ `ue0d2FNRGAP0dRopq40qU`

✓ `ue0b2FNRGAP0dRopq40qU`

✓

`He_just.kept_talking.in.one.long_incredibly.unbroken.`

What do I call you?

Use descriptive names and be consistent!

GOOD `raw_data`, `corpus.cleaned`, `items`

OK `dat`, `data`, `model`, `ratings`, `corpus`

BAD `d`, `df`, `var`, `aaa`, `foo`, `temp`

- ➔ reduce effort
- ➔ stay comprehensible and meaningful (good science is reproducible!)
- ➔ easy to remember and self-explanatory
- ➔ length doesn't matter (much): use **TAB** to autocomplete

Debugging for beginners

Debugging Steps

- 👁 Read the error message
- ❓ Clarify the problem
- 💡 Figure out the logic behind the code
- 📖 Review the code line by line
- ✂ Test on a small sample
- >_ Simplify the code
- 🔍 Google the error message or problem
- 💬 Ask the experts
- 📌 Pinpoint the issue
- ✍ Implement the change
- 😊 Rejoice at your success

Read the error message



Discard your fear. Don't take it personal. It probably tells you **what** went wrong and **where**.

```
Error in read_csv(moses.csv) : could not find function  
"read_csv"
```

```
Error: object 'moses.csv' not found
```

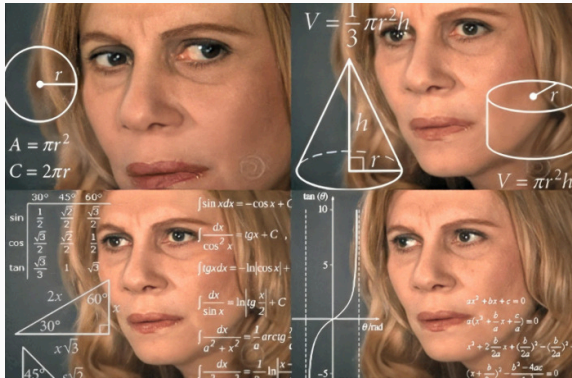
```
Error in select(moses, ID) : unused argument (ID)
```

Clarify the problem

What should have happened? What happened?



Figure out the logic behind the code



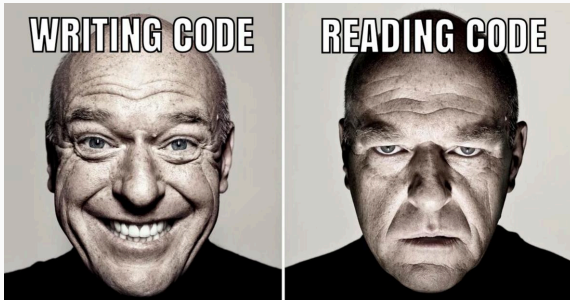
What is the bigger picture?

How does the structure look like?

Cause and effect: What leads/contributes to what?

→ Read the manual/documentation

Review the code line by line



Check for obvious typos:

📁 wrong working directory

📦 package not loaded

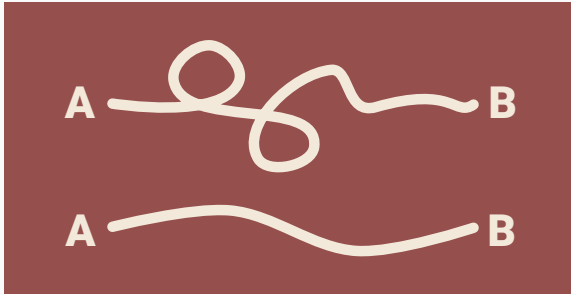
A wrong variable, file, column name, parenthesis

“ missing parenthesis, quotation marks, comma, (semi)colon

Test on a small sample

- ✖ items, 80 fillers, 4 lists, 60 participants, 2000 rows, 50 columns, ...
- ✂ Use a few test cases first.
- 🧩 Don't run the whole script; split it into chunks.
- 📈 Gradually increase the size of your test cases.
- 📦 Leave the whole data set for the last test(s).

Simplify the code

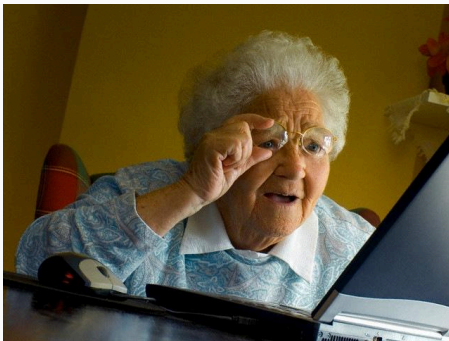


Test the elements separately until they work.

Copy the code for only one element.

Make minimal changes incrementally.

Google error message or general problem



Click through the first 3–5 hits that seem to apply.

“Error in read_csv : could not find function ”read_csv””

→ Example

“R object not found”

→ Example

“dplyr Error in select() : unused argument”

→ Example

Don't “ask to ask”, just ask.

BAD

Hi, I have a problem with R. Can anyone help me?

GOOD

Hi, I have a problem with R. When I run my code I get the following error message: “...”. I tried this and that, but instead of A I get B. Here is a screenshot, a minimal reproducible example, and my session info. Can anyone help me?

Minimal Reproducible Example

It is a small, self-contained piece of code that demonstrates a bug or issue without unnecessary complexity. It helps others quickly understand the problem and suggest solutions.

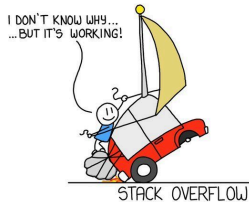
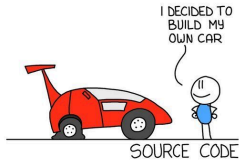
- **Minimal:** Include only the code necessary to reproduce the issue.
- **Complete:** Provide all parts needed to run the code (functions, variables, imports, `sessionInfo()`, OS).
- **Reproducible:** Running the code shows the same error every time.

Get guidance: Wikipedia, StackOverflow, Posit R specific.

Read The Fucking Manual (RTFM)

It probably has all the answers.

(BUG) FIXING CAR USING...



Don't get discouraged



**99 little bugs in the code.
99 little bugs in the code.
Take one down, patch it around.**

127 little bugs in the code...

Getting help

A little stuck: In R

🤔 `help("NAME")`

🤔 `?NAME`

😓 `??NAME`

🏠 RStudio help window

📁 Set the correct working directory `setwd(DIRECTORY)`

⊕ Load the right packages `library(PACKAGE)`

⊖ Unload conflicting packages `detach(PACKAGE)`

🏷️ Disambiguate functions
`dplyr::filter()` or `stats::filter()`

🔄 Clean environment and restart

A little stuck: Online

 Google the exact error message:

“Error in esquisser() : could not find function “esquisser””

 RDocumentation www.rdocumentation.org

 Cheatsheets www.rstudio.com/resources/cheatsheets

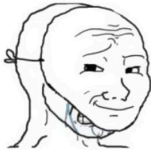
 Tidyverse www.tidyverse.org

 Stackoverflow stackoverflow.com

 Reddit reddit.com/r/rstats

 Prystopska (2025) slides

DESIGNERS



Look, we have similar ideas.



No! You stole my idea.

PROGRAMMERS



Man, I stole your code.



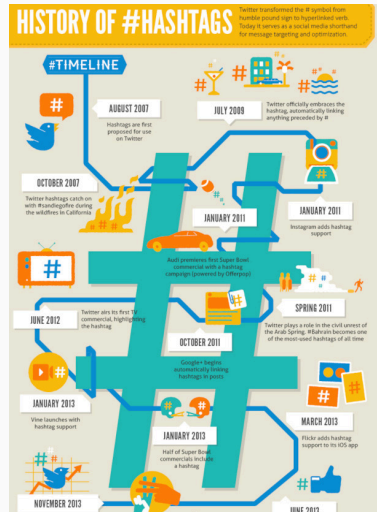
It's not my code.

Data visualization

What is data visualization?

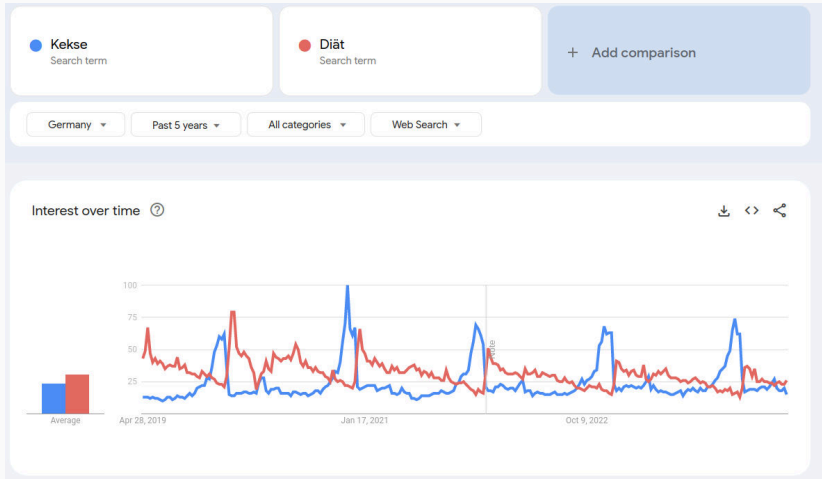


Interdisciplinary field

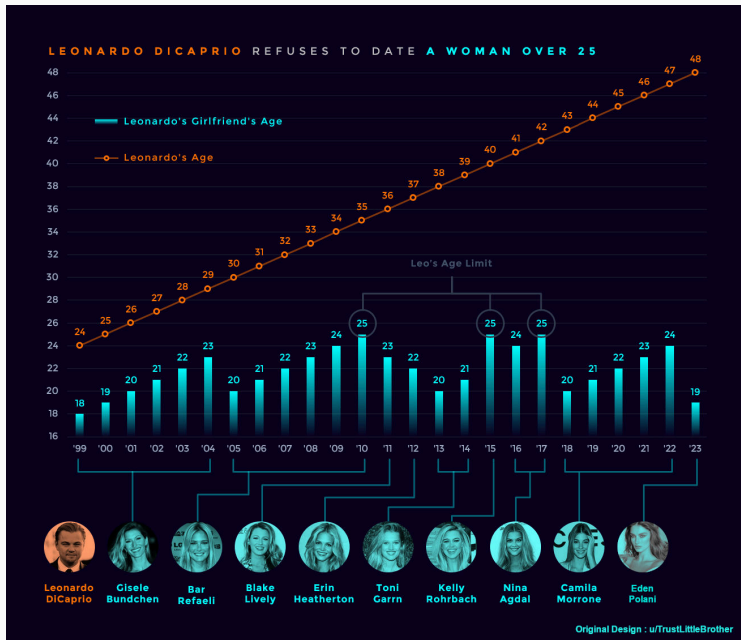


Graphical representation of data & information

What is data visualization?

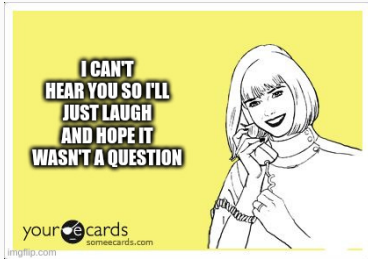


Efficient way of communicating (Google Trends cookies vs. diet)

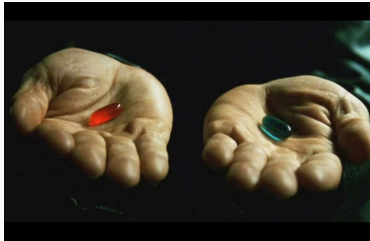


Communicate complex relationships & insights in a comprehensible way

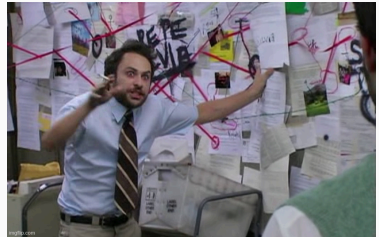
Data visualization goals



Communication & understanding



Decision making



Analysis & exploration

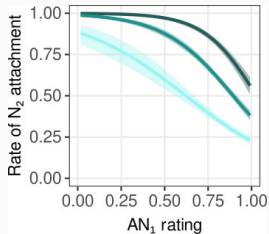


It's likely required in your career

Visualization types

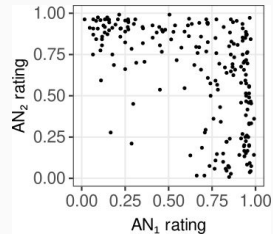
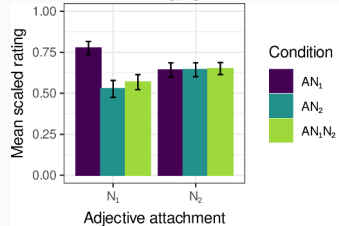
Tables

item	sentence	N1	N2	unclear
3	sommerliche Arbeitskleidung	p 20	p 20	p
11	silberne Bahnhofsuhr	p 20	p 20	p
12	amtliche Baugenehmigung	p 19	p 20	!
16	hohler Bienenstock	p 20	p 20	p
19	gefüttertes Brillenetui	p 20	p 20	p
22	lackiertes Bücherregal	p 20	p 20	p
23	evangelischer Büroangestellter	p 20	p 20	p

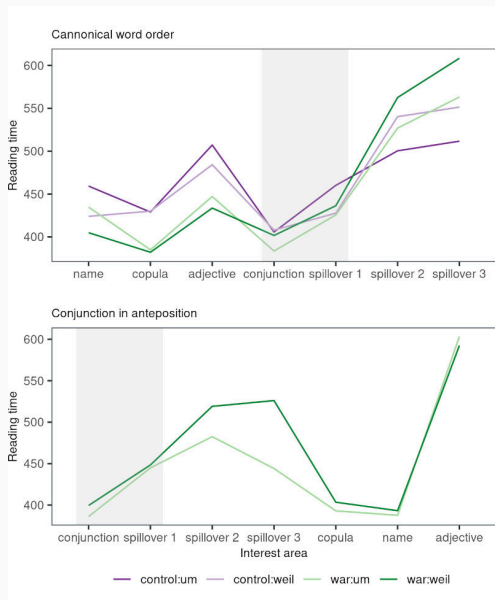


Lines

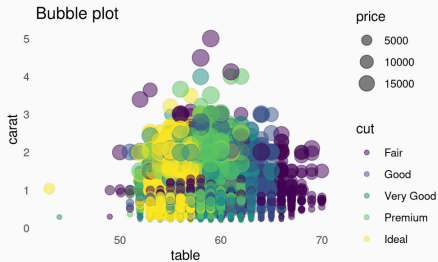
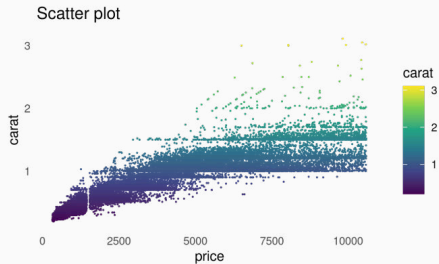
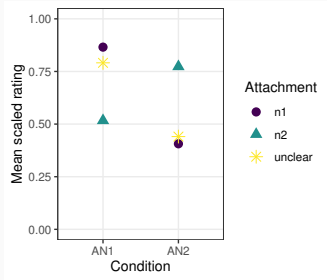
Bars



Points



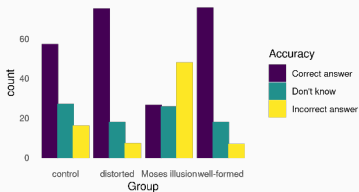
Points



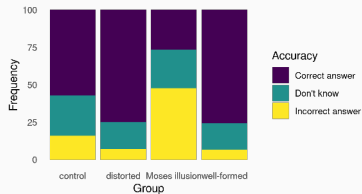
→ <https://r-graph-gallery.com/>

Bars

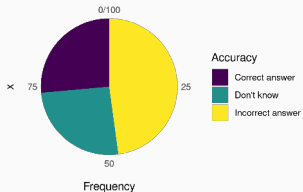
Bar plot



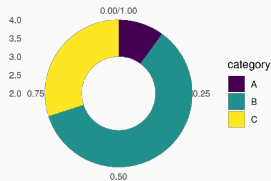
Stacked bar plot



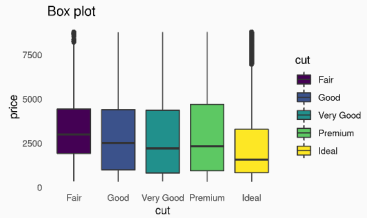
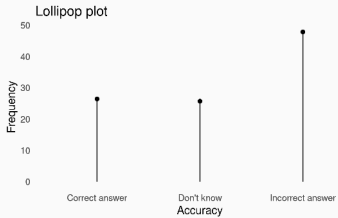
Pie plot



Doughnut plot

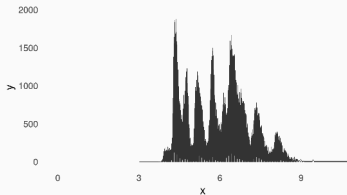


Hybrids

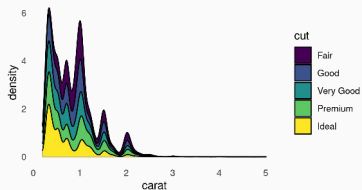


Hybrids

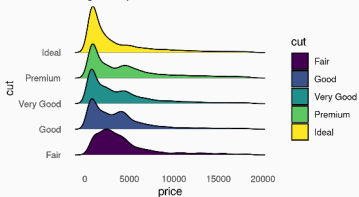
Area plot



Density plot



Ridgeline plot



Violin plot

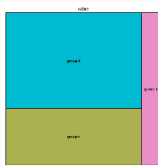


Others

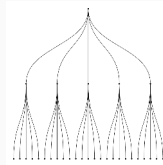
Radar



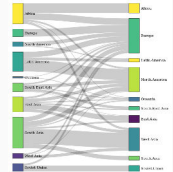
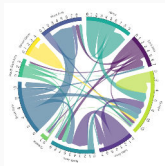
Treemap



Dendrogram



Wordcloud



Chord

Network

Arc

Sankey

Source: www.data-to-viz.com

Accessibility

Principles of Accessibility: Web Content Accessibility Guidelines

Perceivable

Information and interface components are perceivable by users (understandable and not hidden from any senses).

Operable

User interface components and navigation should be operable (usable and cannot require tasks beyond users' abilities).

Understandable

Information and the operation of user interfaces must be understandable (understand the information and how to interact with the interface).

Robust

Content must be so robust that most users can interpret it (as technology progresses, content should remain accessible, including via assistive technologies).

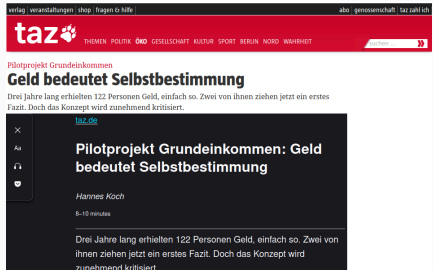
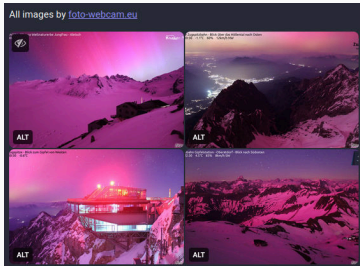


Perceivable Application

Provide **text alternatives** for non-text content (e.g. ALT, subtitles) that can be changed into LARGE PRINT, Braille, speech, translations, simple language.

Create content that can be modified **without losing information or structure** (e.g. simple layout, reader view).

Chose a form that makes **seeing and hearing content easy** (e.g. foreground vs. colors).



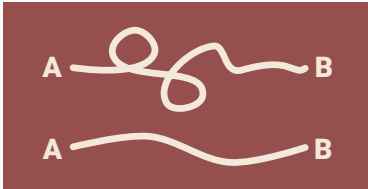
Operable Application

Make all functionality **keyboard accessible** and allow **different input modalities**.

Give **enough time** to read and use content.

Don't design content in a way that is known to cause **seizures or physical reactions**.

Make content **easy to find and navigate**.



Understandable Application

Provide labels and instructions.

Make content readable and understandable.

Make your content appear and operate in predictable ways.

Provide input assistance (to avoid/correct mistakes).



Robust Application

Maximize **compatibility** with current and future software and hardware.

Make compatible with **assistive technologies**.

Simple UI > complex UI.

Use **simple markup** languages.

Test content for crashes.

Include **meta information** (session info).



Wrap-up

Summary

- ✓ getting programming help
- ✓ debugging
- ✓ data visualization goals
- ✓ accessibility and WCOG
- ✓ plot types and choice of visualization
- ▶▶ More data visualization and **ggplot2** (bring your laptop)
- 💡 Recommended reading for troubleshooting:
www.r4wrds.com/intro/m_troubleshooting.html

Homework assignment due May 19th 12:00

Submit 1 R script.

- ❓ Complete assignment 6 (→ ILIAS)
- 💡 “Penguin wrangling **dplyr**” tutorial by Allison Horst:
<https://allisonhorst.github.io/posts/2021-02-08-dplyr-learnr/>
- 💡 Cookbook for R: <http://www.cookbook-r.com/>
- 💡 R Graph Gallery: <https://r-graph-gallery.com/>
- 💡 Tidyverse documentation:
<https://ggplot2.tidyverse.org/>
- 💡 Wickham et al. (n.d.)

References

-  Prysłowska, Anna (Mar. 17, 2025). *Debugging for Beginners*. Lecture slides, Human Sentence Processing (WiSe 2024/25).
-  Wickham, Hadley, Danielle Navarro, and Thomas Lin Pedersen (n.d.). *ggplot2: Elegant Graphics for Data Analysis*. URL: <https://ggplot2-book.org/>.